Technische Hochschule Augsburg

STATISTICS

International Information Systems, 3rd semester

(1) Introduction to Statistics

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STATISTICS TO MAKE SENSE OF THE WORLD

A data-informed present and future

We use statistics to ***get close to the truth*** about the phenomena we are interested in

We start by ***"making guesses"***

- "This neighborhood is richer than this other one"
- "Consumption of margarine increases the risks of hearth disease"
- "Babies who are breastfed are healthier than babies who are bottle-fed"

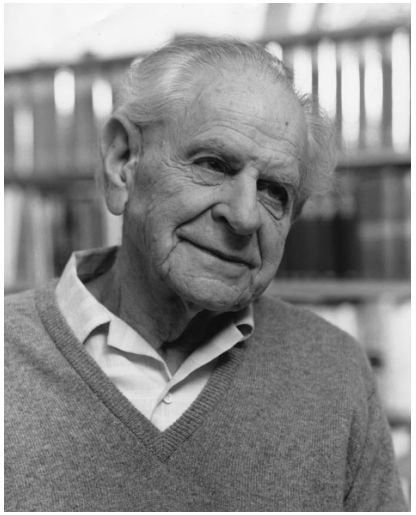
Let's imagine you have a "guess" about the world (e.g. gluten is bad for you)

"***If it disagrees with experience, the guess is wrong***.
In that simple statement is the key to science.
It doesn't matter how beautiful your guess is or
how smart you are or what your name is"

How do we check if our guess is "right" or if it disagrees with experience?

R. Feynmann

TODAY'S TOPICS AND LEARNING GOALS



Hypotheses and Variables

- formulate research questions as ***testable hypotheses*** with ***dependent*** and ***independent variables***
- identify possible **proxies** for a variable

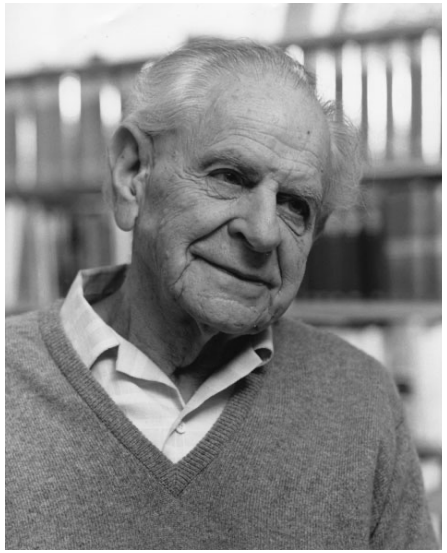
Experimental vs. Correlational research

- understand and explain the difference between ***correlational*** and ***experimental research***

Scales of Measurement

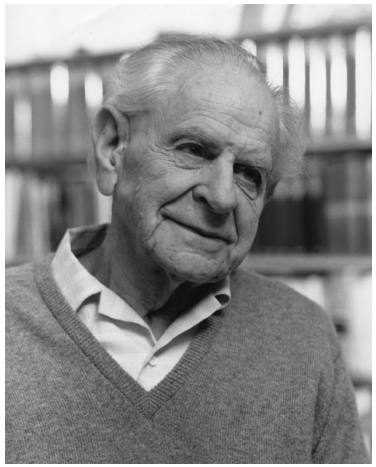
- distinguish between ***categorical*** and ***continuous variables***
- identify the appropriate ***scale of measurement*** for a variable

HYPOTHESES AND VARIABLES



HOW DO WE KNOW IF OUR GUESS DISAGREES WITH EXPERIENCE?

From Claim to Testable Hypothesis



K. Popper

"Breastfeeding is the best choice"

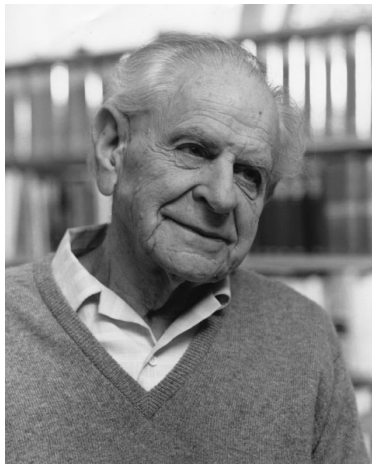
Claim
a "guess" about the state of the world

"Breastfed children are more intelligent than bottle-fed children"

"IQ of breastfed children is higher than IQ of bottle-fed children"

Hypothesis
a **testable statement** about the state of the world to be tested with reference to empirical evidence

TESTING A HYPOTHESIS



- We do not look for evidence **confirming** our hypotheses,
  but we try to look for evidence **disproving** our hypotheses. **Why?**

- **Confirming a Hypothesis:**

I have a hypothesis:
"I believe covid vaccines are dangerous"

I look for evidence confirming it
"Every time a death is reported in my community,
I asked if the person was vaccinated"

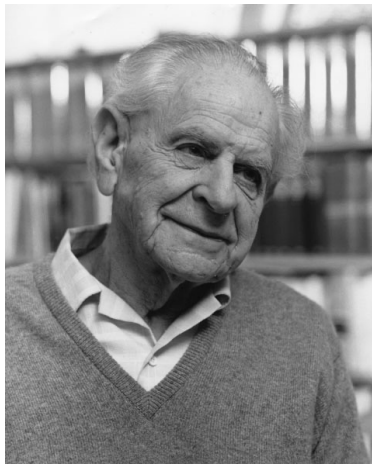
I find it
"As in my community lots of people got the vaccine,
it's likely that if a death is reported, that person was vaccinated"

~~**BINGO: My hypothesis is confirmed!**~~

Pseudo-science!!!

- **This is a case of confirmation bias:**
  it's easy for us to find **confirming** evidence,
  but this **does not** prove our hypothesis!

TESTING A HYPOTHESIS



- We do not look for evidence **confirming** our hypotheses, but we try to look for evidence **disproving** our hypotheses. **Why?**

- **Disproving a Hypothesis:**

I have a hypothesis → **I look for evidence which would falsify it**

- **I find it** → **I reject my hypothesis**
- **I fail to find it** → **I fail to reject the hypothesis ... until someone else does**

- **Scientific hypotheses can always be disconfirmed**

HYPOTHESIS TESTING AND VARIABLES

What are we measuring?

- We measure **variables** to test our hypotheses
- Most hypotheses can be expressed in terms of two variables, e.g.

"Consumption of margarine increases the risks of health disease"

proposed cause (known) ***proposed outcome (unknown)***

independent variable
predictor variable

dependent variable
outcome variable

HYPOTHESIS TESTING AND VARIABLES

What are we measuring?

- We measure **variables** to test our hypotheses
- Most hypotheses can be expressed in terms of two variables, e.g.

"Eating fast food causes stress"

proposed cause (independent variable) ***proposed outcome (dependent variable)***

How do we measure the variables "eating fast food" and "stress"?

HYPOTHESIS TESTING AND VARIABLES

Proxies

- When we can't measure something directly, we need a proxy, a variable **we can measure**

Proxy
a variable that serves in place of an unobservable or immeasurable variable

"Eating fast food causes stress"

proposed cause ***proposed outcome***

- for "eating fast food": e.g. surveys, habit logs
  (can we trust what people say in a survey?)
- for stress: e.g. heart rate, self-reporting
  (is heart rate a good way to measure stress?)

Choosing the right variables and good proxies for them is part of your job as an analyst / researcher

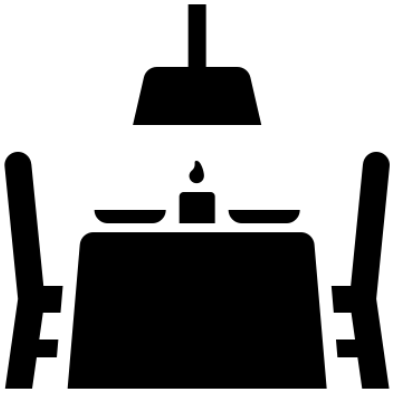
- **No "magical AI / software" which will do this for you!**

Questions?

Howell, D. C. (2016). *Fundamental statistics for the behavioral sciences*. Cengage Learning.

EXERCISE

Hypotheses and Variables

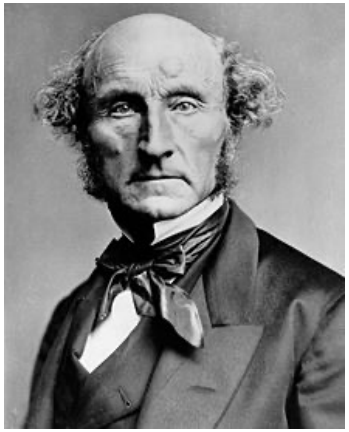


A study about the social social desirability of behavior
(Pliner and Chaiken, 1990)

- They examined the amount of food eaten by male and female participants in the presence of a person of the same gender, or a person of the opposite gender
  - What is the dependent variable in that study?
  - What are the independent variables?
  - What would you expect was the experimenter's hypothesis?

Discuss with your neighbour (5 minutes)

EXPERIMENTAL VS. CORRELATIONAL RESEARCH



EXPERIMENTAL AND CORRELATIONAL RESEARCH

“Eating fast food causes stress”

proposed cause ***proposed outcome***

- Most research questions can be broken down into a ***proposed cause*** and a ***proposed outcome***
- The implication may be more or less subtle, or stated more or less explicitly
- How do we infer ***cause*** and ***effect***?

CORRELATIONAL RESEARCH

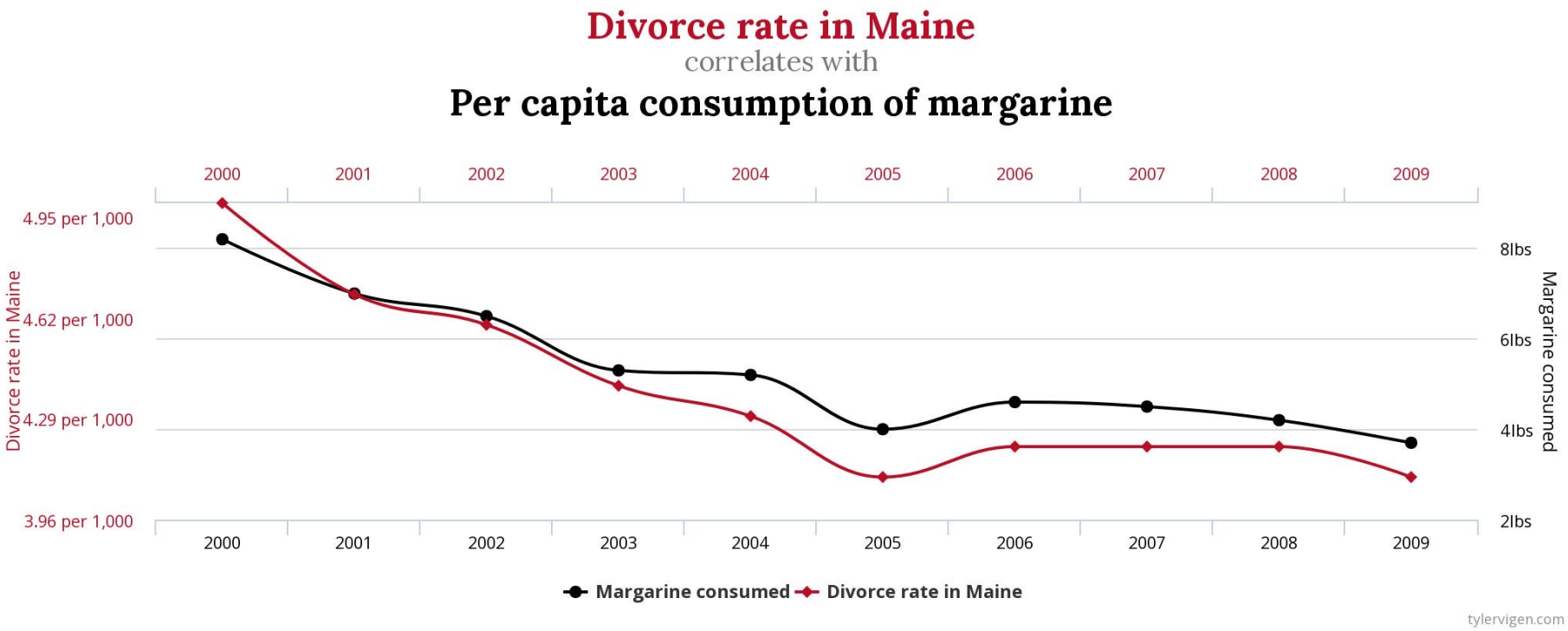
“Eating fast food happens together with stress”

proposed cause ***proposed outcome***

- In correlational studies, variables are often measured ***simultaneously***
- **we can not draw conclusions about causality**
  - which of the variable “comes first”?
  - can one happen without the other?
  - can there be a **confound**, a third element causing both?

https://tylervigen.com/

CORRELATIONAL RESEARCH

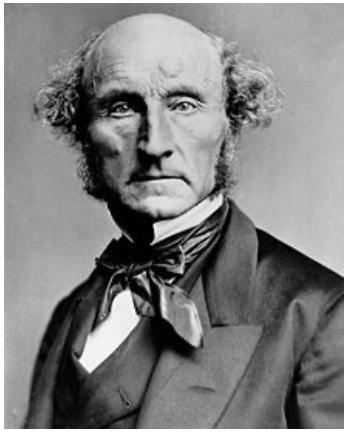


Correlation is not Causation!

FROM CORRELATION TO CAUSATION

David Hume (1739): **to infer that A (cause) causes B (effect)**

- A and B must occur close together in time
- A must occur before B
- A should never occur without B



- John Stuart Mill (1865):
  you also need **to rule out other explanations** for B (confounds)
  - without A, you shouldn't observe B

EXPERIMENTAL RESEARCH

"Eating fast food happens together with stress"

proposed cause ***proposed outcome***

- Experimental research requires creating and comparing two (or more) controlled situations
  - one in which the cause is present (e.g. *people are given fast food*)
  - one in which the cause is absent (e.g. *people are given food that's not fast food*)
- E.g. one group receives a drug, the other a placebo, with random assignment of participants to groups
- This manipulation allows us to ***infer causation***

Experiment
a procedure comparing two or more controlled situations which only differ with regards to a proposed cause to study a proposed outcome

Questions?

SCALES OF MEASUREMENT

CATEGORICAL VS CONTINUOUS VARIABLES

Variables represent ***properties*** of entities, which can vary regarding to

- People (e.g. height, eye color, IQ, reading speed)
- Locations (e.g. precipitation, unemployment rates), Time (e.g., profit, number of cancerous cells)
- Experimental manipulation (e.g. one group gets a drug, one group gets a placebo)

Categorical or **discrete variables** can take only a few possible values

grammatical gender, voted party, eye color

- **Binary variables** can take only one out of two values

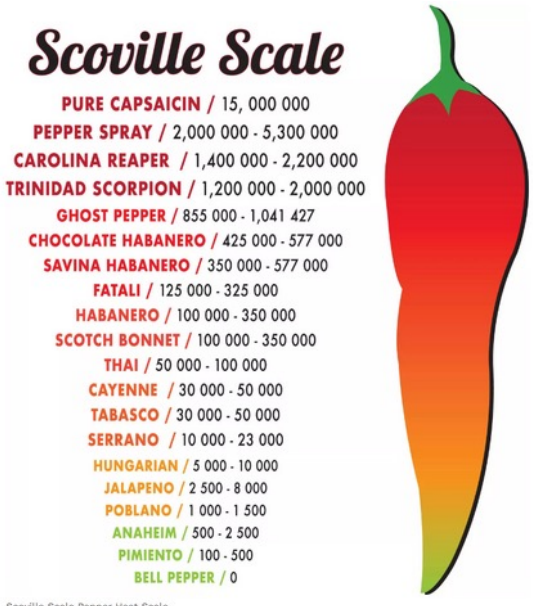
positive / negative to covid, treatment / placebo, married / not married

Continuous variables can assume any value between the highest and lowest point in a scale

speed, hormone levels in blood, amount of rain

VARIABLES AND MEASUREMENT

- Measurement means to **assign *numbers* to *objects* to assess a *property* of the object**
  - Height
  - Stress (using heart rate as a proxy)
  - Press Freedom Index of a country
  - Scoville Scale of spiciness of hot peppers
- Depending on what we measure and how we measure it, these **numbers** may have different **properties**
  - **Scales of measurement** are the ways these numbers relate to each other



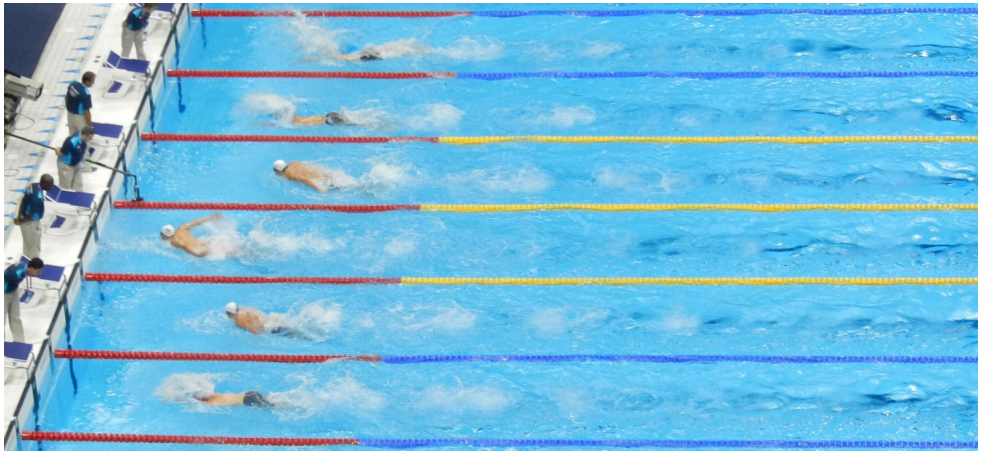
SCALES OF MEASUREMENT

↗ Knowing the **scale of measurement** you are using will help you perform the right analysis

What was your bib number?

"18"

label



Where did you finish in the competition?

"18"

rank

How many seconds did it take you?

"18"

continuous measure

SCALES OF MEASUREMENT

The nominal scale

We use the **nominal scale** to label / categorize entities

- one entity can only be assigned to one label
- if we use a number, it's used only a label, to distinguish an entity from the other (e.g. numbers on football jerseys)

- We use the nominal scale for ***categorical or discrete variables***

  grammatical gender, eye color, voted party

  - including ***binary variables***

    positive / negative to covid, treatment / placebo, married / not married

SCALES OF MEASUREMENT

The ordinal scale

- What if the categories can be **ordered** along a continuum or **ranked**?

We use the **ordinal scale** to assigns objects to one value / category in a set **if the values can be ordered**

- we can express the order using numbers
- but we **don't know how distant** the categories are from each other

language proficiency, education

Likert Scale ("On a scale from 1 to 5... ")

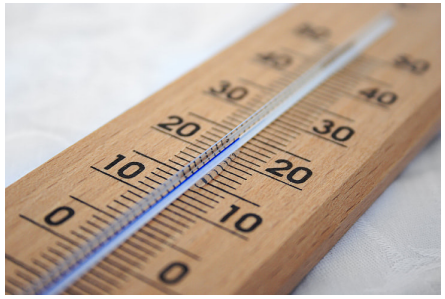
SCALES OF MEASUREMENT

The interval and the ratio scale

The **interval scale** is used when we can speak of **regular differences** between the points in the scale

- ↗ **equal intervals – equal differences**

Fahrenheit or Celsius scale of temperature



The **ratio scale** is used when we can also speak meaningfully of **ratios** between the points in the scale

- ↗ because there is a **(non-arbitrary) zero point**

Kelvin scale of temperature, time, length, weight

EXERCISE

Scales of measurement

Imagine we are describing a swimming competition you took part in
What scale of measurement should be used to describe...?

- **your bib number?**
  “26”
  categorical variable
  nominal scale
- **where you finished in the competition?**
  “26”
  categorical variable
  ordinal scale
- **how cold the water was?**
  “26” (C)
  continuous variable
  interval scale
- **how many seconds it took you?**
  “26”
  continuous variable
  ratio scale

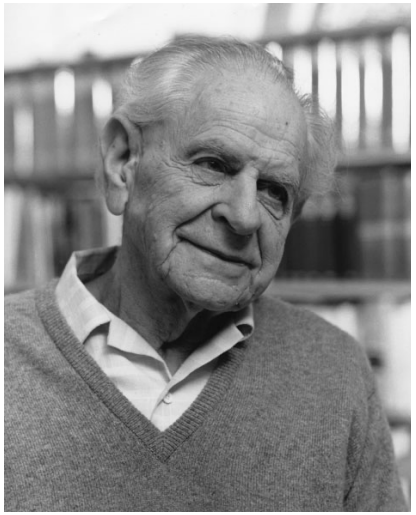
EXERCISE

Scales of measurement

Imagine you go to the cinema
What scale of measurement should be used to describe...?

- **the movie genre?**
  (drama, comedy, adventure, documentary, etc.)
  categorical variable
  nominal scale
- **the movie duration?**
  (e.g. 120 minutes)
  continuous variable
  ratio scale
- **the movie ratings?**
  (e.g. very poor, poor, good, very good)
  categorical variable
  ordinal scale

TODAY'S TOPICS AND LEARNING GOALS



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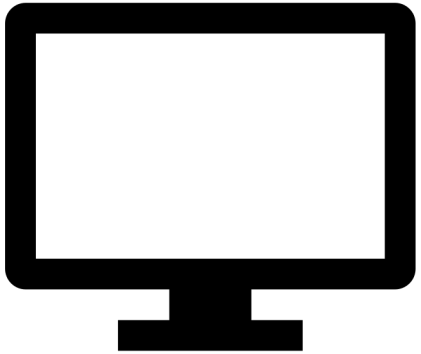
EXERCISE

A UX research team wants to understand if their new design for a web site helps users find the information they are looking for more quickly.
A group of users is given a list of information to search on the old version and on the new version of the web site. The researchers are recording how many times the participants click on a html link during the search before finding the information they were looking for.



- **Identify the dependent variable and the independent variable**

- **What type of variables are the dependent variable and the independent variable and on what scale of measurement are they measured?**

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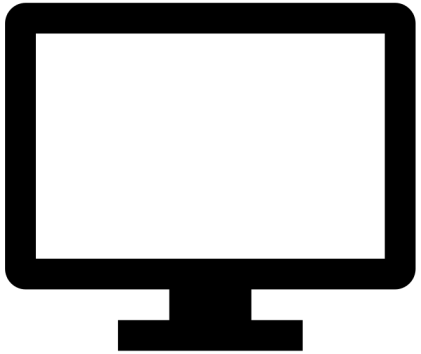


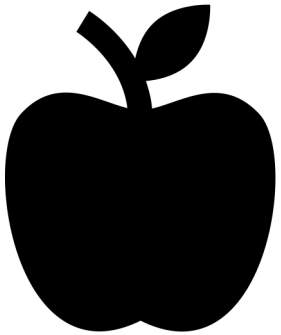
- **Identify the dependent variable and the independent variable**

  DV: number of clicks on html links
  IV: website design

- **What type of variables are the dependent variable and the independent variable and on what scale of measurement are they measured?**

  DV: continuous variable, ratio scale
  IV: categorical variable (new / old design)

EXERCISE



A team of researchers wants to study if following a vegan diet improves overall cardiovascular health. They recruit 22 pairs of twins and give one twin from each pair with either a vegan or omnivore diet (both diets were healthy). After eight weeks, the researchers measure the level of "bad" cholesterol in the participants' blood (measured as mg of LDL-C per dL).

- **Identify the dependent variable and the independent variable**

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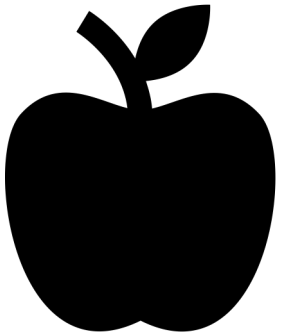
EXERCISE



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- **Identify the dependent variable and the independent variable**

  DV: levels of bad cholesterol in the blood
  IV: type of diet

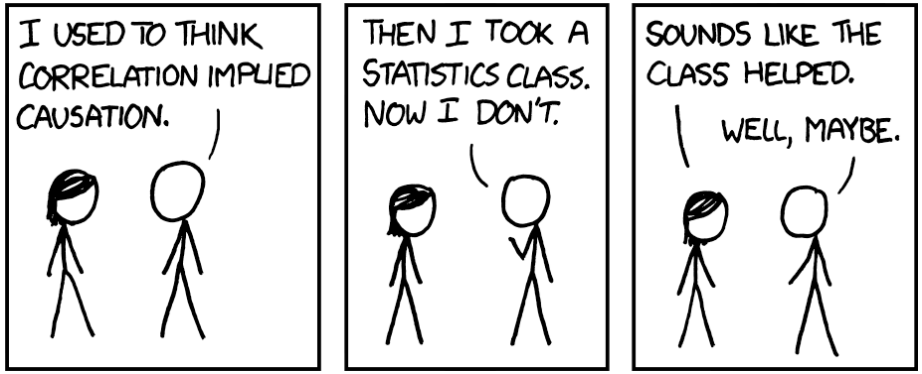
- **What type of variables are the dependent variable and the independent variable and on what scale of measurement are they measured?**

  DV: continuous variable, ratio scale
  IV: categorical variable (vegan / omnivore diet)

Questions?

SUMMING UP

xkcd



- Statistics is the science of **learning from data**
- To get **close to the truth** about a phenomenon
  we start by formulating **testable hypotheses** about the world
  - in terms of relations **between variables** (dependent vs. independent)
  - **falsifiable** (we look for evidence **disconfirming** the hypothesis)
- **Correlational studies:** variables are often measured simultaneously – we can not draw conclusions about causality
- **Experimental studies** create two (or more) controlled situations to ***infer causation***
- Variables can be categorical or continuous and can be measured using different **scales of measurement** (**nominal**, **ordinal**, **interval**, **ratio**)